

Bansilal Ramnath Agarwal Charitable Trust's
Vishwakarma Institute of Technology, Pune-37

(An autonomous Institute of Savitribai Phule Pune University)



Department of Multidisciplinary
Engineering

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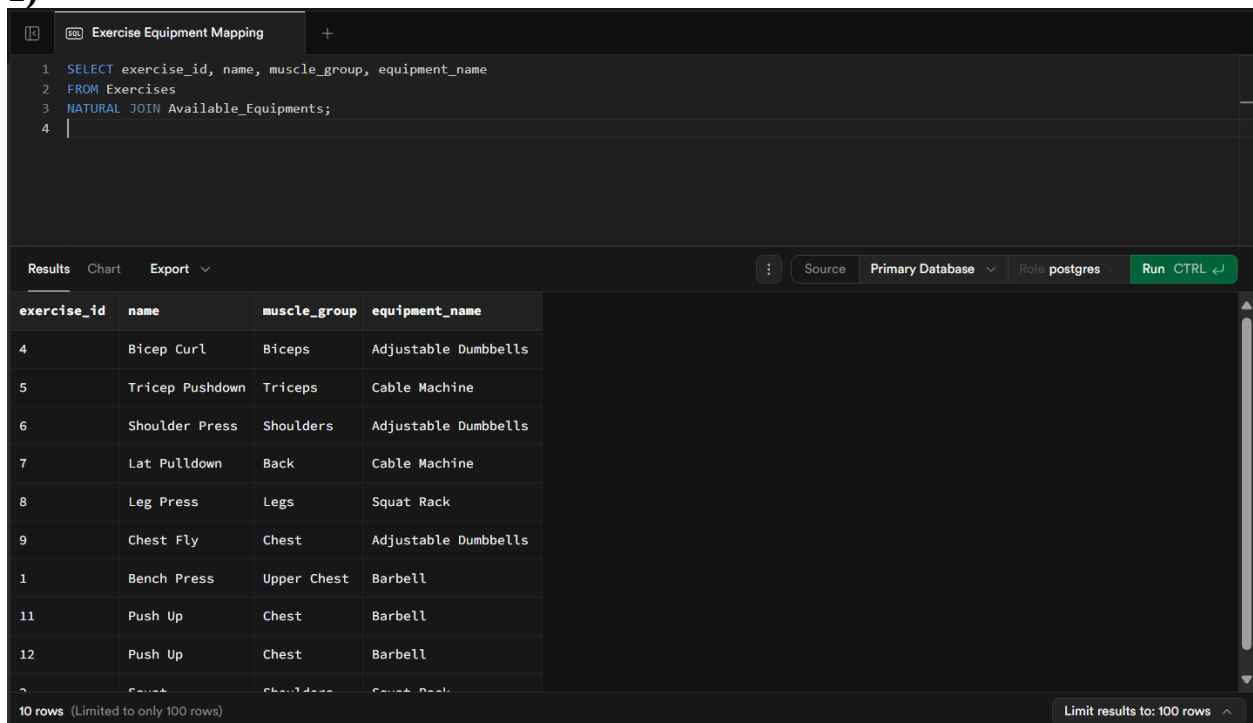
Title:

1. Natural join
2. Theta join
3. Equi join
4. Left outer
5. Right outer
6. Full outer union
7. Self join

Description: In this assignment, we applied different SQL joins on the Exercises and Available_Equipments tables of our workout planner database. We used Natural Join to automatically match common columns, Theta Join with conditions, and Equi Join for equality-based matching. We also performed Left Outer, Right Outer, and Full Outer Joins to include unmatched rows from one or both tables. Lastly, a Self Join was used on the Exercises table to compare records within the same table. This helped us understand how joins combine data from multiple tables to extract meaningful results.

Screenshots/Output:

1)



The screenshot shows a SQL IDE window titled "Exercise Equipment Mapping". The query editor contains the following SQL code:

```
1 SELECT exercise_id, name, muscle_group, equipment_name
2 FROM Exercises
3 NATURAL JOIN Available_Equipments;
4
```

The results pane displays a table with the following data:

exercise_id	name	muscle_group	equipment_name
4	Bicep Curl	Biceps	Adjustable Dumbbells
5	Tricep Pushdown	Triceps	Cable Machine
6	Shoulder Press	Shoulders	Adjustable Dumbbells
7	Lat Pulldown	Back	Cable Machine
8	Leg Press	Legs	Squat Rack
9	Chest Fly	Chest	Adjustable Dumbbells
1	Bench Press	Upper Chest	Barbell
11	Push Up	Chest	Barbell
12	Push Up	Chest	Barbell
10	Squat	Shoulders	Squat Rack

The interface includes tabs for "Results", "Chart", and "Export". The "Results" tab is active, showing the query results. The "Export" tab is also visible. The "Run" button is labeled "Run CTRL ↵". The status bar at the bottom indicates "10 rows (Limited to only 100 rows)" and "Limit results to: 100 rows".

2)

Exercise Equipment MappingExercise Equipment Mismatch+

```
1 SELECT e.name, eq.equipment_name
2 FROM Exercises e
3 JOIN Available_Equipments eq
4 ON e.equipment_id > eq.equipment_id;
5
```

ResultsChartExportSourcePrimary DatabaseRole postgresRun CTRL ↵

name	equipment_name
Bicep Curl	Barbell
Bicep Curl	Squat Rack
Bicep Curl	Cable Machine
Tricep Pushdown	Barbell
Tricep Pushdown	Squat Rack
Shoulder Press	Barbell
Shoulder Press	Squat Rack
Shoulder Press	Cable Machine
Lat Pulldown	Barbell
Lat Pulldown	Squat Rack

15 rows (Limited to only 100 rows)Limit results to: 100 rows ^

3)

Exercise Equipment MappingExercise Equipment MismatchExercise Equipment Mapping+

```
1 SELECT e.exercise_id, e.name, eq.equipment_name
2 FROM Exercises e, Available_Equipments eq
3 WHERE e.equipment_id = eq.equipment_id;
4
```

ResultsChartExportSourcePrimary DatabaseRole postgresRun CTRL ↵

exercise_id	name	equipment_name
4	Bicep Curl	Adjustable Dumbbells
5	Tricep Pushdown	Cable Machine
6	Shoulder Press	Adjustable Dumbbells
7	Lat Pulldown	Cable Machine
8	Leg Press	Squat Rack
9	Chest Fly	Adjustable Dumbbells
1	Bench Press	Barbell
11	Push Up	Barbell
12	Push Up	Barbell
3	Squat	Squat Rack

10 rows (Limited to only 100 rows)Limit results to: 100 rows ^

4)

The screenshot shows a database query editor with a tab labeled "Exercise Equipment Mapping". The query is a SQL LEFT JOIN between the "Exercises" table and the "Available_Equipments" table. The results table has three columns: "exercise_id", "name", and "equipment_name". It displays 10 rows of data, showing exercises 4 through 12 and their corresponding equipment. The interface includes a "Run" button and a "Limit results to: 100 rows" option.

```
1 SELECT e.exercise_id, e.name, eq.equipment_name
2 FROM Exercises e
3 LEFT JOIN Available_Equipments eq
4 ON e.equipment_id = eq.equipment_id;
5
```

exercise_id	name	equipment_name
4	Bicep Curl	Adjustable Dumbbells
5	Tricep Pushdown	Cable Machine
6	Shoulder Press	Adjustable Dumbbells
7	Lat Pulldown	Cable Machine
8	Leg Press	Squat Rack
9	Chest Fly	Adjustable Dumbbells
1	Bench Press	Barbell
11	Push Up	Barbell
12	Push Up	Barbell
~	Squat	Squat Rack

10 rows (Limited to only 100 rows) Limit results to: 100 rows ^

5)

The screenshot shows a database query editor with a tab labeled "Equipment and Associate". The query is a SQL RIGHT JOIN between the "Exercises" table and the "Available_Equipments" table. The results table has three columns: "exercise_id", "name", and "equipment_name". It displays 15 rows of data, showing exercises 4 through 12 and their corresponding equipment. The interface includes a "Run" button and a "Limit results to: 100 rows" option.

```
1 SELECT e.exercise_id, e.name, eq.equipment_name
2 FROM Exercises e
3 RIGHT JOIN Available_Equipments eq
4 ON e.equipment_id = eq.equipment_id;
5
```

exercise_id	name	equipment_name
4	Bicep Curl	Adjustable Dumbbells
5	Tricep Pushdown	Cable Machine
6	Shoulder Press	Adjustable Dumbbells
7	Lat Pulldown	Cable Machine
8	Leg Press	Squat Rack
9	Chest Fly	Adjustable Dumbbells
1	Bench Press	Barbell
11	Push Up	Barbell
12	Push Up	Barbell
~	Squat	Squat Rack

15 rows (Limited to only 100 rows) Limit results to: 100 rows ^

6)

Exercise Equipment Mapping		
<pre> 1 SELECT e.exercise_id, e.name, eq.equipment_name 2 FROM Exercises e 3 FULL OUTER JOIN Available_Equipments eq 4 ON e.equipment_id = eq.equipment_id; 5 </pre>		
exercise_id	name	equipment_name
4	Bicep Curl	Adjustable Dumbbells
5	Tricep Pushdown	Cable Machine
6	Shoulder Press	Adjustable Dumbbells
7	Lat Pulldown	Cable Machine
8	Leg Press	Squat Rack
9	Chest Fly	Adjustable Dumbbells
1	Bench Press	Barbell
11	Push Up	Barbell
12	Push Up	Barbell
2	Squat	Squat Rack
15 rows (Limited to only 100 rows)		

7)

Exercise Equipment Mapping		
<pre> 1 SELECT e1.name AS Exercise1, e2.name AS Exercise2, e1.equipment_id 2 FROM Exercises e1 3 JOIN Exercises e2 4 ON e1.equipment_id = e2.equipment_id 5 AND e1.exercise_id <> e2.exercise_id; 6 </pre>		
exercise1	exercise2	equipment_id
Bicep Curl	Chest Fly	4
Bicep Curl	Shoulder Press	4
Tricep Pushdown	Lat Pulldown	3
Shoulder Press	Chest Fly	4
Shoulder Press	Bicep Curl	4
Lat Pulldown	Tricep Pushdown	3
Leg Press	Squat	2
Chest Fly	Shoulder Press	4
Chest Fly	Bicep Curl	4
Bench Press	Push Up	1
Push Up	Bench Press	1
16 rows (Limited to only 100 rows)		